

Highlights

Optimization of a Recycling Mixer-Reactor-Clarifier Activated Sludge System Using a Physically-Constrained Statistical Surrogate

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- One surrogate represents the complete recycling N -stage activated sludge system.
- One 115-variable NLP jointly optimizes five controls and 110 physical states.
- Ten Clarifier-layer solids states make the SRT inventory physically traceable.
- A full-rank scaled quadratic basis gives one fixed multiresponse fit.
- One exact dynamic replay validates each selected multistart local candidate.

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ABSTRACT

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